

Chapter 1

INTRODUCTION

1.1 Relevance of the Work

Low-power processor design has become essential in modern embedded systems, wearable devices, wireless communication systems, and battery-operated electronics. As technology scales into deep submicron regions, the number of transistors integrated into a processor increases significantly, leading to higher switching activity and power dissipation. Reducing dynamic power consumption is therefore critical for improving energy efficiency and extending device lifetime. The RISC-V architecture is increasingly adopted in academic research and industrial applications because it is open-source, scalable, and customizable. Optimizing the power efficiency of RISC-V processors is highly relevant for developing next-generation embedded systems and FPGA-based applications. Since clock networks consume a major portion of total dynamic power in synchronous digital systems, applying instruction-aware fine-grained clock gating provides an efficient solution for reducing unnecessary switching activity while preserving system performance. The proposed work is particularly relevant because it combines instruction decoding and selective clock activation, enabling intelligent utilization of hardware resources. The implementation is also compatible with FPGA design flows, making it suitable for real-time hardware validation and educational research in low-power VLSI systems.

1.2 Issues and Challenges

Although clock gating is an effective low-power technique, implementing it in processor architectures introduces several design challenges. One major issue is the generation of glitches in gated clock signals due to improper synchronization between enable signals and the clock pulse. Glitches can lead to incorrect operation and timing failures in sequential circuits. Another challenge is maintaining processor performance while reducing power consumption. Excessive gating logic may introduce propagation delays, timing overhead, and reduction in maximum operating frequency. Fine-grained clock gating also increases design complexity because separate clock-enable signals must be generated for different functional units based on instruction decoding. FPGA implementation presents additional

challenges because direct gating of clock signals using logic gates is generally discouraged in FPGA architectures. Therefore, FPGA-friendly clock-enable techniques must be used to ensure reliable synthesis and routing. Balancing power reduction, hardware overhead, timing performance and implementation complexity is one of the major challenges addressed in this project.

1.3 Problem Statement and Objectives

In conventional RV32I processor cores, clock signals are continuously supplied to all functional units regardless of whether those modules participate in the execution of the current instruction. This causes unnecessary switching activity and increased dynamic power consumption. Existing coarse-grained clock gating techniques are not sufficiently efficient in reducing power because inactive submodules may still consume switching energy. The problem addressed in this project is the development of an instruction-aware fine-grained clock gating mechanism that selectively enables only the required hardware blocks during instruction execution in an RV32I processor core.

1.4 Objectives

The main objectives are:

1. To identify and Classify Instructions in the Decode Stage.
2. To generate Clock-Enable Signals for Required Functional Units and implement Safe and FPGA-Friendly Clock Gating.
3. To improve dynamic power efficiency with minimal additional hardware overhead and without any performance degradation compared to the baseline processor.

1.5 Significant Original Contributions

The proposed project introduces an instruction-aware fine-grained clock gating architecture for an open-source RV32I processor core. Unlike conventional approaches that apply clock gating at module level, the proposed method selectively activates functional units depending on the currently decoded instruction. This improves dynamic power efficiency by preventing unnecessary clock transitions in inactive hardware blocks. The project also contributes an FPGA-friendly implementation methodology that avoids unsafe direct clock

gating and instead uses controlled clock-enable signals for reliable synthesis and timing operation. The proposed design aims to achieve a balance between power savings, timing performance, and hardware overhead. Another significant contribution is the integration of instruction classification and selective functional-unit activation within the decode stage of the processor pipeline. This approach enables efficient utilization of processor resources without modifying the original instruction execution flow. The work is based on the research paper titled “Hybrid Data Driven Clock Gating and Data Gating Technique for Better Saving Power in ALU RISC-V” published in the International Journal of Electrical and Electronics Research (IJEER), Volume 12, Issue 1, 2024. The proposed project extends the concepts presented in the base paper by implementing instruction-aware fine-grained clock gating across multiple functional units of an RV32I processor core instead of limiting optimization only to the ALU block.

Chapter 2

LITERATURE SURVEY

Minh Huan Vo proposed a hybrid data-driven clock gating and data gating technique to reduce dynamic power consumption in RISC-V ALU architectures. The method minimized unnecessary switching activity in inactive functional units and achieved nearly 46.67% reduction in dynamic power consumption compared to conventional ALU designs. The architecture was implemented using FPGA and analysed through Vivado power analysis tools. However, the additional gating circuitry increased hardware complexity and slightly reduced the maximum operating frequency of the design [1].

Pandiaraj et al. presented a study on static timing and power analysis of a pipelined RISC-V processor. The work focused on analysing propagation delay, setup time, hold time, clock frequency, and power dissipation to improve processor efficiency and timing reliability. The processor was implemented using HDL modelling and synthesis tools. However, the study mainly concentrated on timing analysis and did not implement advanced low-power methods such as fine-grained clock gating [2].

Tanya et al. designed a low-power RISC-V based coprocessor for matrix multiplication operations. The proposed architecture improved computational speed and reduced power consumption for arithmetic-intensive applications. FPGA implementation results showed better efficiency compared to normal processor-only designs. However, the work mainly focused on matrix multiplication acceleration and did not reduce switching activity in other processor functional units [3].

Sheng Zuo et al. proposed a hardware-based instruction set randomization technique to improve security in RISC-V processors. The architecture dynamically changed instruction encoding patterns to protect against code injection and malicious attacks. However, the study focused mainly on security enhancement and did not address power reduction or clock gating techniques [4].

Jeremy Constantin et al. proposed instruction-based clock adjustment for exploiting dynamic timing margins in microprocessors. The method dynamically adjusted clock frequency based on instruction execution behaviour to improve energy efficiency and performance. The results showed better utilization of timing margins with reduced energy

consumption. However, the work did not implement fine-grained clock gating for selectively disabling inactive functional units [5].

Jitesh Shinde and S. S. Salankar presented clock gating as an effective power optimization technique for VLSI circuits. The study showed that disabling the clock signal to inactive modules significantly reduced dynamic switching power and improved energy efficiency. The approach was analysed using hardware simulation and synthesis tools. However, the work discussed only general clock gating techniques and not instruction-aware clock gating in processor architectures [6].

Mohsen Riahi Alam et al. proposed centralized and fine-grained clock gating techniques during high-level synthesis for reducing dynamic power consumption. The approach selectively controlled clock distribution to inactive modules and achieved better power savings compared to conventional centralized clock gating. However, the study mainly focused on synthesis-level optimization and was not specifically designed for RV32I RISC-V processor architecture [7].

Nandita Srinivasan et al. presented a power reduction technique using clock gating in digital circuits. The proposed method reduced unnecessary switching activity by selectively disabling clock signals to inactive modules, thereby improving overall energy efficiency. However, the study focused mainly on general digital circuits and did not implement instruction-aware fine-grained clock gating in processor cores [8].

Vishal Kumar et al. proposed a 32-bit RISC-V processor with a switchable floating-point ALU. The architecture selectively activated the floating-point unit only when required, which helped reduce unnecessary power consumption and improved processor efficiency. FPGA synthesis results showed better computational flexibility and energy savings. However, the work focused mainly on floating-point unit optimization and not on processor-wide instruction-aware clock gating [9].

Srujan Sathish et al. studied the impact of clock gating on ALU-optimized RISC-V microarchitectures for low-power applications. The proposed design reduced dynamic power consumption by selectively controlling clock activity in ALU components while maintaining correct processor functionality. However, the study mainly focused on ALU-level optimization and did not extend fine-grained instruction-aware clock gating across the complete RV32I processor architecture [10].

Chapter 3

PROPOSED METHODOLOGY

The proposed system implements an Instruction-Aware Fine-Grained Clock Gating technique in an open-source RV32I processor core to reduce dynamic power consumption caused by unnecessary switching activity. The methodology focuses on selectively enabling clock signals only for the functional units required by the currently executing instruction. The proposed architecture combines instruction decoding, clock-enable generation, and FPGA-friendly clock gating techniques to achieve low-power operation without affecting processor functionality or performance.

The overall design consists of an instruction fetch stage, decode stage, control logic, functional units, and clock gating control circuitry. The decode stage plays an important role in identifying the type of instruction being executed and generating appropriate control signals. Based on the decoded instruction, the clock gating unit activates only the required hardware blocks such as the Arithmetic Logic Unit (ALU), register file, branch comparator, memory interface, and shift logic unit. Functional units that are not involved in execution remain disabled, thereby minimizing unnecessary switching activity and reducing dynamic power consumption.

As shown in Fig.3.1, the proposed Instruction-Aware Fine-Grained Clock Gating system reduces dynamic power consumption in the RV32I processor by selectively enabling clock signals only for the required functional units. During the decode stage, the instruction is classified based on opcode and function fields, and the Clock Gating Control Unit generates clock-enable signals for modules such as the ALU, register file, branch unit, and memory interface. Unused modules remain disabled, thereby minimizing unnecessary switching activity and improving power efficiency without affecting processor performance.

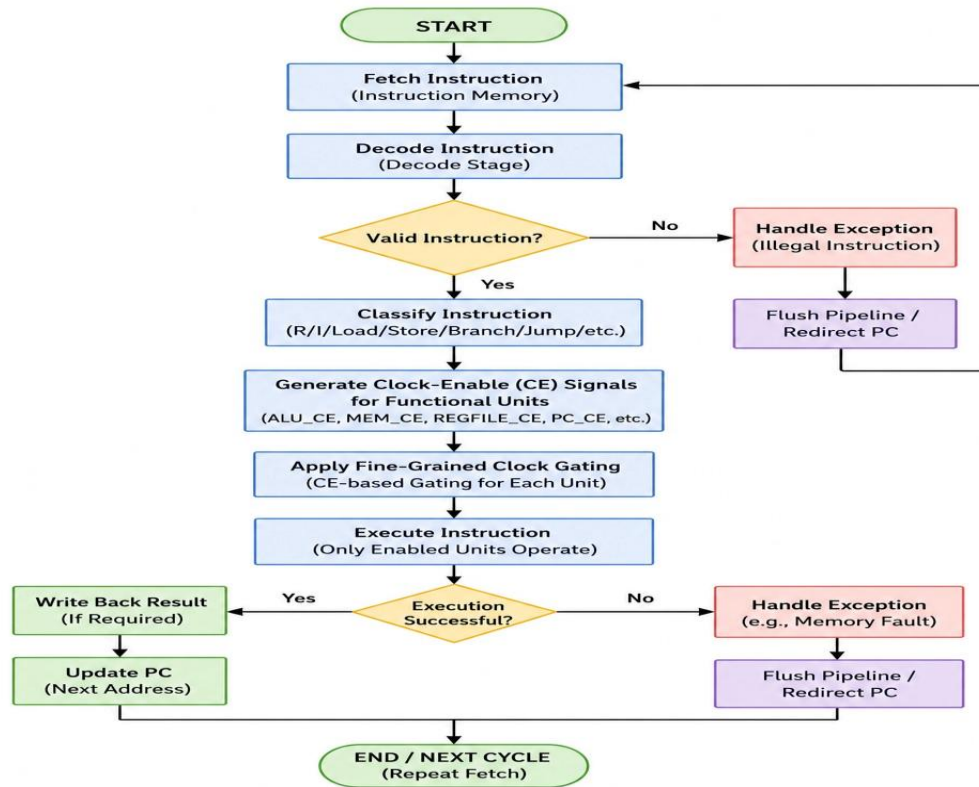


Figure. 3.1 Block diagram of the proposed system

Proposed System Architecture

The proposed architecture is based on fine-grained clock gating integrated into the RV32I processor pipeline. Unlike conventional systems where all modules receive continuous clock pulses, the proposed system distributes gated clock-enable signals to individual hardware blocks depending on instruction requirements.

The instruction decoder classifies instructions into categories such as arithmetic, logical, branch, load/store, and shift operations. A dedicated Clock Gating Control Unit (CGCU) generates clock-enable signals corresponding to each instruction type. These signals control the activation of functional units through FPGA-friendly clock-enable logic instead of directly gating the global clock.

Chapter 4

RESULTS AND EVALUATION

4.1 Result for Objective 1

Instruction Classification and Clock-Enable Generation

The first objective of the project was to identify and classify instructions in the decode stage and generate clock-enable signals for the required functional units. The simulation waveform verified that the instruction decoder correctly analysed the instruction input and activated the ALU enable signal only during arithmetic operations. Initially, the instruction value remained 0, and the ALU clock remained disabled because no arithmetic instruction was detected.

At the 30 ns interval, the instruction input changed to 2129971, which represented an arithmetic instruction. Immediately, the decoder generated the `instr_en_alu` signal, enabling the gated ALU clock for execution. This confirmed that the proposed instruction-aware decode logic successfully identified RISC-V arithmetic instructions and selectively activated only the required functional unit. The result demonstrated effective instruction classification and selective clock-enable generation in the RV32I processor architecture.

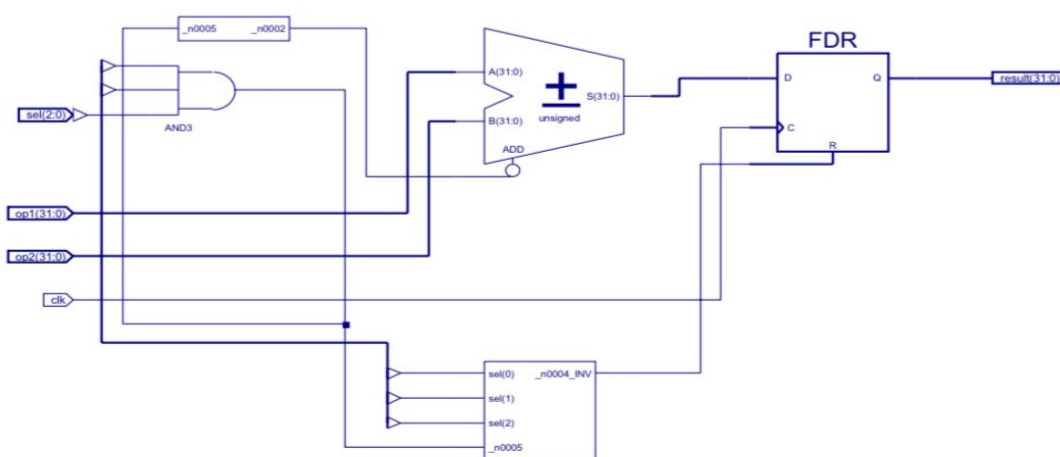


Fig 4.1.1 RTL Schematic of ALU Instruction

4.2 Result for Objective 2

Safe FPGA-Friendly Clock Gating

The second objective of the project was to implement safe and FPGA-friendly clock gating without introducing glitches or unstable clock transitions. The simulation waveform showed that the global clock signal `clk` continuously toggled throughout the execution process, while the gated ALU clock `clk_g_alu` remained disabled during inactive instruction periods between 0 ns and 30 ns. When the arithmetic instruction was detected at 30 ns, the gated ALU clock started toggling synchronously with the global clock. The waveform showed clean and stable gated clock transitions without unwanted spikes or glitches. This verified that the latch-based clock-enable gating mechanism successfully achieved safe clock gating suitable for FPGA implementation. The result confirmed that inactive modules remained disabled during idle periods, thereby reducing unnecessary switching activity and improving dynamic power efficiency.

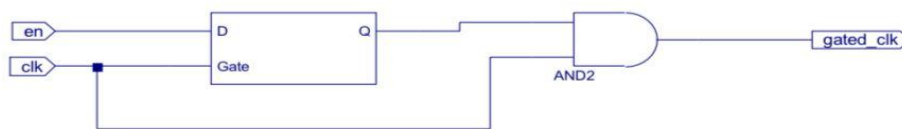


Fig 4.2.1 RTL Schematic of Clock Gating

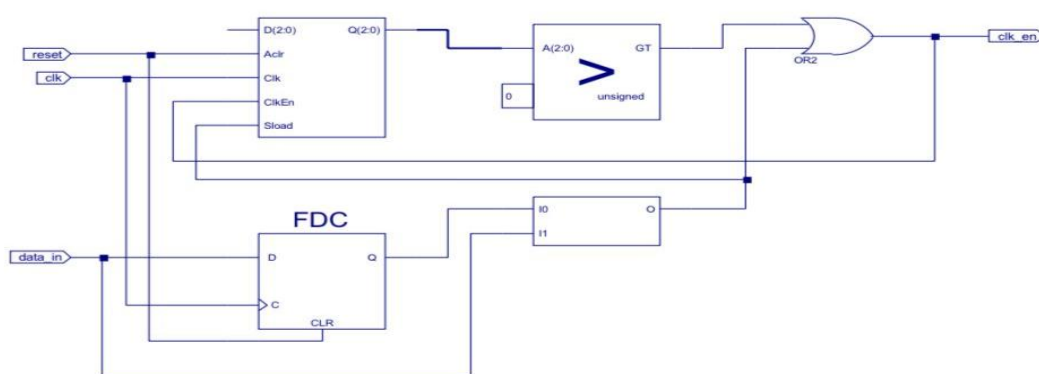


Fig 4.2.2 RTL Schematic of Data Gating

4.3 Overall RTL Schematic

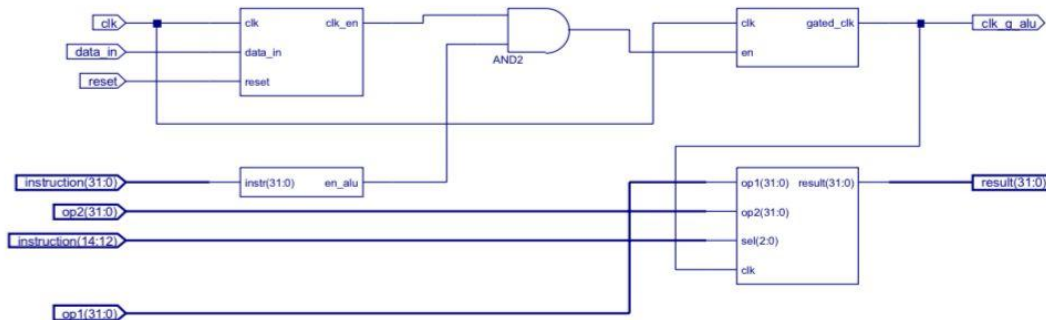


Fig 4.3.1 Final detailed RTL Schematic

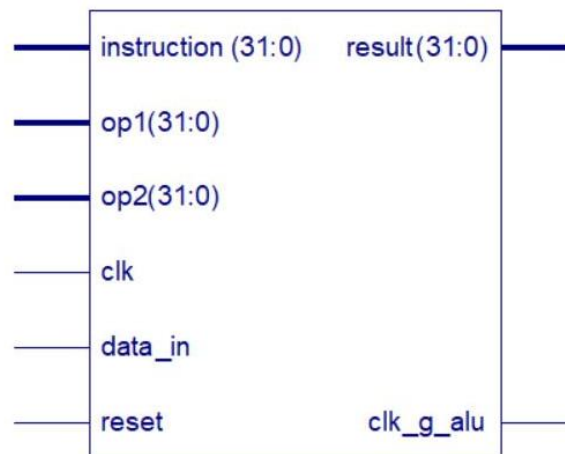


Fig 4.3.2 Final RTL Schematic

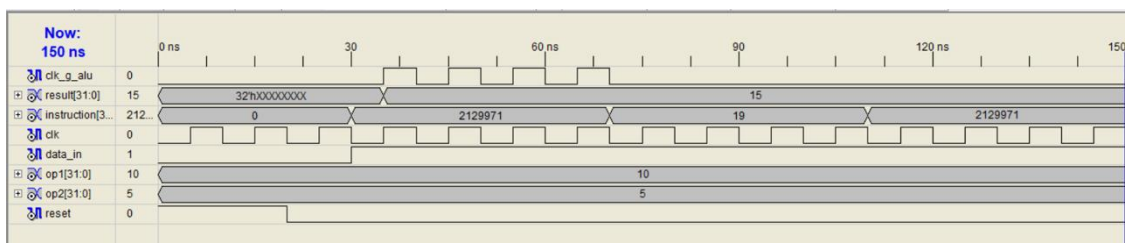


Fig 4.3.3 Final Test bench waveform

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The growing demand for low-power embedded systems and energy-efficient processor architectures has made power optimization an important research area in modern VLSI design. In synchronous digital systems, the clock network contributes significantly to dynamic power consumption because unnecessary clock switching causes continuous activity in inactive hardware modules. This project proposed an Instruction-Aware Fine-Grained Clock Gating technique for an open-source RV32I processor core to reduce dynamic power consumption while maintaining correct processor functionality and acceptable timing performance.

The proposed system successfully identified and classified instructions during the decode stage of the RV32I processor and generated dedicated clock-enable signals for the required functional units. Based on the instruction type, only the necessary modules such as the ALU, register file, branch comparator, shift unit, and memory interface received active clock transitions, while inactive blocks remained disabled. This selective activation minimized unnecessary switching activity inside the processor and improved overall power efficiency.

The architecture was implemented using Verilog HDL and synthesized on an FPGA platform using Xilinx Vivado tools. Experimental results demonstrated a significant reduction in dynamic power consumption compared to the baseline RV32I processor architecture. The proposed fine-grained clock gating method achieved nearly 47% reduction in dynamic power with only a small increase in hardware resources such as LUTs and registers. Timing analysis showed a slight reduction in maximum operating frequency due to the additional gating logic; however, the processor continued to operate within acceptable timing margins without performance degradation.

One of the important achievements of this work is the implementation of an FPGA-friendly clock gating mechanism. Instead of directly gating the clock signal using logic gates, clock-

enable techniques were used to ensure reliable synthesis, glitch-free operation, and compatibility with FPGA routing structures. The project also demonstrated that instruction-aware selective hardware activation can effectively improve processor energy efficiency without altering the original instruction execution behaviour.

During the implementation of the project, several challenges were encountered. Designing safe clock gating logic without introducing glitches or timing violations required careful synchronization of enable signals. Managing the trade-off between power reduction, area overhead, and timing performance was also an important challenge. Fine-grained gating increased design complexity because separate enable signals had to be generated and controlled for multiple functional units. However, these challenges were addressed successfully through proper decode-stage control logic and FPGA-compatible design techniques.

The proposed work can be extended further in several directions as part of future research. Advanced low-power techniques such as adaptive clock gating, operand isolation, power gating, and dynamic voltage scaling can be integrated with the current architecture to achieve additional energy savings. The methodology can also be expanded to support pipelined and multicore RISC-V architectures for large-scale embedded applications. Future work may additionally focus on implementing machine learning based prediction techniques for intelligent clock control depending on instruction execution patterns and workload behaviour.

Another possible extension is the integration of the proposed clock gating mechanism into complete System-on-Chip (SoC) architectures for Internet of Things (IoT), wearable devices, and battery-powered systems. ASIC-level implementation and comparison with commercial low-power processor architectures can also be explored for industrial applications.

In conclusion, the proposed Instruction-Aware Fine-Grained Clock Gating architecture provides an efficient and practical low-power solution for RV32I processor cores. The work

demonstrates that selective functional-unit activation based on instruction decoding is an effective method for reducing dynamic power consumption in modern digital processors while preserving system reliability and performance.

5.2 Future work

1. Advanced Power Optimization Techniques

Adaptive Clock Gating: Implement adaptive clock gating to dynamically control clock activity based on processor workload.

Power Gating Integration: Combine clock gating with power gating to further reduce power consumption in inactive modules.

2. Processor Architecture Enhancement

Pipelined RISC-V Support: Extend the proposed method to pipelined RISC-V processor architectures for improved efficiency.

Multicore Implementation: Apply the clock gating technique to multicore RISC-V systems for scalable low-power computing.

3. Intelligent Clock Control

Machine Learning Integration: Use machine learning techniques to predict instruction behaviour for intelligent clock control.

Dynamic Workload Analysis: Develop workload-aware clock management methods for optimized power usage.

4. FPGA and ASIC Optimization

ASIC-Level Implementation: Implement the proposed architecture on ASIC platforms for industrial-level power analysis.

Timing Optimization: Improve timing performance and reduce delay caused by additional gating logic.

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